

Minwoo Park

mwpark@engineering.upenn.edu

linkedin.com/in/minwoo-park1 | github.com/minpark830 | minpark830.github.io

Summary

Electrical engineer (M.S., University of Pennsylvania, Dec 2026) working across embedded software, controls, and hardware: C++/Python motion control at Varian, FreeRTOS firmware and Kalman filter localization for a drone swarm, and PCB schematic, layout, and compliance testing at Stryker R&D.

Education

University of Pennsylvania

Expected Dec 2026

Master of Science in Electrical Engineering, Concentration in AI; GPA: 3.62/4.00

California Polytechnic State University, San Luis Obispo

Sep 2021 – Jun 2025

Bachelor of Science in Electrical Engineering, Minor in Computer Science; GPA: 3.96/4.00

Dean's List (Fall 2021 – Spring 2025); President's Honor List (2021 – 2025)

Skills

Programming: C, C++, Python (matplotlib), MATLAB, Simulink, SystemVerilog, RISC-V assembly, Java, JavaScript, SQL, LabVIEW, ROS

Embedded: FreeRTOS, STM32 (Arm Cortex-M), interrupts, I2C, SPI, UART, Arduino, Raspberry Pi, FPGA (AMD Xilinx Vivado)

Guidance, navigation, and control (GN&C): PID and state-space control, Kalman filtering (EKF), state estimation, formation control, trajectory generation, path planning, convex optimization

Hardware and test: Altium Designer (schematic capture, PCB layout, footprints, bill of materials), Eagle, LTspice/SPIICE, mixed-signal circuits, soldering, Fusion 360, oscilloscope, spectrum and network analyzers, multimeter, EMC and power compliance testing (IEC 60601)

Networking and tools: WebSockets, FastAPI, peer-to-peer radio, Linux, Git, GitHub Actions (CI/CD)

Languages: English, Korean, Spanish (limited)

Work Experience

Varian (Siemens Healthineers) — Palo Alto, CA

May 2026 – Present

Embedded Software Engineer / Control Systems Intern

- Built servo control, optimization, and path and treatment-planning algorithms in C++ and Python for radiotherapy machine systems, improving servoing performance and motion accuracy.
- Increased trajectory generation speed by 30% on average by optimizing the planning algorithms.

Kumar Lab, University of Pennsylvania — Philadelphia, PA

Aug 2025 – Present

Research Assistant (avg. 10 hrs/week)

- Developing a research paper on multi-agent payload control, including literature synthesis, problem formulation, and analysis of trajectory generation and controller performance.
- Built a Crazyflie quadrotor test framework for reproducible trajectory and controller experiments.

CPSquare (Cal Poly Cyber-Physical Systems Lab) — San Luis Obispo, CA

Sep 2024 – Jun 2025

Research Assistant (avg. 10 hrs/week)

- Co-developed dynamic leader-follower formation control (triangle and line formations) for a 3-drone Crazyflie swarm coordinated over peer-to-peer radio (DTR token ring).
- Wrote the leader drone's firmware app in C as FreeRTOS tasks, and a Python FastAPI ground server with a WebSocket link to each drone (one Crazyradio 2.0 per drone).
- Developed EKF localization fusing optical-flow and time-of-flight range sensors on each drone.
- Built failure detection and recovery mechanisms robust to variable environmental and sensor conditions.
- Co-authored *Framework for Multi-Agent Coordination and Distributed Localization in Micro-UAVs*.

Cal Poly Department of Electrical Engineering — San Luis Obispo, CA Jan 2025 – Jun 2025
Instructional Teaching Assistant (avg. 15 hrs/week)

- Graded and gave feedback on EE 302 (Classical Controls) and EE 342 (Classical Control Systems Lab).

Stryker (Endoscopy) — San Jose, CA Jun 2024 – Sep 2024
Advanced Operations Electrical Engineering Intern

- Built LabVIEW automation for the 1788 camera head line, cutting manufacturing time by up to 20%.
- Adapted Precision Light Source calibration software to the L11 line, saving 5% of floor space.
- Revised the 1788 quality-check software, improving reliability and cutting debugging time by 2+ weeks.

Cal Poly Department of Electrical Engineering — San Luis Obispo, CA Oct 2023 – Jun 2024
Instructional Teaching Assistant (avg. 10 hrs/week)

- Taught soldering, PCB design, and microcontroller fundamentals in EE 143 (Circuit Analysis Lab).
- Graded assignments for EE 113, EE 228, and EE 342.

Stryker (Interventional Spine) — Portage, MI Jun 2023 – Sep 2023
R&D Electrical Engineering Intern

- Designed the schematic and PCB layout of an Arduino-integrated test board in Altium Designer, reducing MultiGen 2 splitter box testing time by 75%.
- Built the board's schematic symbols and footprints by hand, chose parts to power and cost requirements, and generated the bill of materials (BOM) for purchasing.
- Tested and verified EMC and power compliance of RF ablation console PCBAs to IEC 60601.
- Authored comprehensive documentation for the MultiGen 2 test board for R&D teams.

Projects

Vision-Controlled Robot Manipulator Sep 2024 – Dec 2024

- Implemented forward/inverse kinematics and computer vision algorithms to control an OpenManipulator-X robot arm for pick-and-place operations.

Ultrasonic Underwater Sensor Network Sep 2024 – Dec 2024

- Built an end-to-end underwater acoustic link carrying DHT11 sensor data through ultrasonic hardware and an analog receive chain (AGC, PLL tone decoder) to a Raspberry Pi web server.

Custom IR RC Car (EE 329 final project) Apr 2024 – Jun 2024

- Built an IR-controlled RC car on an STM32 microcontroller, using I2C and SPI drivers and interrupts to integrate motor drivers, IR and ultrasonic sensors, an accelerometer, and an LCD.

RISC-V 32-bit MCU Jan 2024 – Mar 2024

- Implemented a pipelined RISC-V CPU with cache and peripheral I/O in SystemVerilog on an FPGA.

4-Bit DAC PCBA Apr 2022 – Jun 2022

- Designed a binary-weighted resistor DAC and integrated it with an Arduino speaker system.

Relevant Coursework

Electrical / Computer Engineering: Circuit Analysis I/II/III, Digital & Computer Design, Assembly Programming, Digital & Continuous Systems, Classical Controls, Digital & Analog Electronics, Microcontroller System Design, Vision-Based Robotics, Optimization Theory, Modern Convex Optimization, Learning in Robotics, Embedded Systems/Devices, and Linear Systems Theory

Computer Science: Data Structures, Object-Oriented Programming/Design, Discrete Structures, Systems Programming, Machine Learning, Machine Learning for Time-Series Data, Principles of Deep Learning, Big Data Analytics, and Statistics for Data Science

Certifications

FCC Amateur Radio Technician License